# RESEARCH PORTFOLIO

## T81: Balanced Ternary Architecture for Deterministic, Auditable AI

A comprehensive research program exploring ternary-native computing substrates, governed deterministic inference, multiplication-free AI, and post-binary systems.

| Core Focus | Primary Stack | Key Result |
|---|---|---|
| **Balanced Ternary Computing** | **T81 Foundation TISC / HanoiVM / Axion** | **T3_K Quantization ~2.63-bit LLM weights** |

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GitHub: github.com/t81dev | Focus: Ternary Computing · Deterministic AI · Systems Research

Portfolio generated from public repositories under github.com/t81dev
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# 1. Research Vision

Modern computing is built almost entirely on binary foundations. While binary has delivered extraordinary scale and reliability, it forces every concept that is naturally three-valued — neutrality, signed magnitude, or symmetric decision spaces — into awkward encodings. The T81 research program investigates whether **balanced ternary** (digits −1, 0, +1) can serve as a first-class computational substrate for domains where determinism, auditability, and structural symmetry matter more than raw binary throughput.

The central thesis is not that ternary should replace binary. Rather, ternary offers a **bounded semantic augmentation**: a disciplined three-valued layer that can reduce structural complexity in neutrality-dominant workloads, enable multiplication-free inference, and provide bit-exact reproducibility guarantees that floating-point binary systems struggle to maintain.

## Core Research Questions

- Can explicit ternary semantics reduce structural complexity without compromising binary determinism?
- What quantifiable advantages (energy, density, determinism) does balanced ternary offer for modern LLM inference?
- How can a runtime enforce policy *before* side effects, producing verifiable decision objects rather than opaque model outputs?
- What hardware and memory architectures become viable when the native digit set is {−1, 0, +1}?
- Can post-binary cryptographic constructions operating natively in ternary space offer new security properties?

# 2. Technical Foundations

## Balanced Ternary

Balanced ternary is a base-3 positional number system using the digit set {–1, 0, +1}. Unlike unbalanced ternary (0, 1, 2), it is symmetric around zero. Every integer has a unique representation with no need for a separate sign bit. Negation is a simple digit-wise flip. The information density is log■(3) ≈ 1.585 bits per trit — higher than binary while retaining signed symmetry that maps naturally onto many numerical and decision problems.

## Why It Matters for AI

Modern neural network weights, after training, often concentrate near zero. Balanced ternary quantization maps weights to {–1, 0, +1}, enabling **multiplication-free** matrix multiplications (conditional addition and subtraction only). This yields both compression (≈2.63 bits per weight for T3_K) and potential energy and latency advantages on appropriate hardware. More importantly, the discrete, truncation-only nature of ternary arithmetic supports **bit-exact reproducibility** — a property floating-point pipelines cannot guarantee across platforms.

## Duotronic Model

The research adopts a **duotronic** stance: binary remains the control and transport substrate; ternary augments it where meaning, neutrality, and symmetry are first-class concerns. The Ternary Logic Unit (TLU) is defined as a coprocessor rather than a wholesale replacement. Formal semantics and evaluation criteria are captured in the Duotronic Whitepaper, kept intentionally upstream of any implementation.

# 3. Flagship Project: T81 Foundation

T81 is a substrate for producing **canonical decision objects with identity** from governed AI tasks. Instead of treating AI output as something to be interpreted after the fact, T81 produces durable, content-addressed decision bundles that carry complete provenance, support deterministic replay, and can be consumed across systems without the original execution environment.

## Architectural Pillars

### TISC ISA (v1.9.0 Frozen)

Deterministic ternary instruction set with fully specified edge-case behavior and bit-exact semantics.

### T81VM / HanoiVM

Deterministic interpreter executing TISC bytecode with Axion safety hooks, overflow protection, and reproducible traces.

### Axion

Governance kernel that evaluates policy *before* side effects. Allow/Deny verdicts are deterministic; denied operations never materialize.

### CanonFS

Immutable, hash-verified storage for decision chains and evidence. Content-addressed bundles provide cryptographic proof of integrity.

### T81Lang

High-level deterministic language frontend compiling to TISC IR and bytecode with effect boundaries and safety guarantees.

## System Guarantees

- Identity by design — content-addressed bundles provide cryptographic proof of decision integrity
- Deterministic replay — identical inputs always produce identical decision objects
- Pre-side-effect policy enforcement — no computation occurs for denied operations
- Cross-system portability — bundles remain consumable without the original runtime
- Complete provenance — full audit trail from input to decision is verifiable

# 4. Ternary Quantization & Model Tooling

A major applied research thrust is the first working balanced ternary quantization path for modern large language models. The T3_K scheme maps weights to three discrete levels, achieving approximately 2.63 bits per weight while preserving competitive perplexity.

## Illustrative Results (Gemma-2-2B-IT)

| Method | Size | WikiText-2 PPL |
|---|---|---|
| FP16 | 4.80 GB | 5.91 |
| Q4_K_M | 2.80 GB | 6.48 |
| **T3_K** | **2.4 GB** | **6.38** |

T3_K achieves ~16% smaller size than Q4_K_M with lower perplexity on the reported model. Entropy analysis of real LLM weights in balanced ternary shows high coding efficiency (Huffman coding approaches theoretical limits).

## Supporting Libraries & Tools

### t81lib

Header-first C++20 + Python library providing balanced ternary arithmetic, packed ternary GEMMs, quantization helpers, and Torch integration.

### ternary

End-to-end T3_K quantization pipeline with safetensors → GGUF conversion, runnable with llama.cpp / Ollama.

### ternary-tools

Rust-based balanced-ternary-aware GGUF inspector and debugger — the 'file(1) of the ternary age'.

### t81-python

High-level Python package with clean APIs, type hints, and integration examples for Hugging Face and llama.cpp.

# 5. Hardware, Memory & Systems Research

Beyond software runtime and quantization, the program maintains active research into ternary-native hardware primitives, memory architectures, and systems software.

### t81-hardware

Verilog/SystemVerilog simulations, testbenches, FPGA targeting, and software emulators of ternary co-processors.

### ternary-fabric

Ternary-native memory and interconnect co-processor designed to accelerate AI and signal processing workloads.

### ternary-memory-research

Systematic evaluation of ternary memory cells and periphery under realistic noise, energy, and latency guardrails; SPICE-backed guard ledger.

### ternary_gcc_plugin

GCC plugin enabling C/C++ to express balanced ternary operations and lower them deterministically into a normative helper ABI.

### ternary-pager

User-space pager experiments for falsifying ternary semantic compression hypotheses.

Current memory research status indicates robust headroom under mismatch and noise sweeps, while comparator handoff remains an open challenge — illustrating the program's commitment to adversarial evaluation and transparent failure reporting.

# 6. Cryptography & Security Experiments

A parallel research line explores cryptographic constructions that operate natively on balanced ternary data rather than mapping ternary into binary primitives.

### trinity

T81 Cipher Suite — four post-binary algorithms (REFC, AECS, RTPE, TRTSC) operating directly in balanced ternary space.

### trinity-pow

Recursive, entropy-pruned balanced ternary mining algorithm targeting SHA-256 leading zeros with maximally random trit distributions.

### trinity-decrypt

Decrypt-focused companion implementations and validation.

These experiments are exploratory. They serve to probe whether ternary-native constructions can yield interesting hardness assumptions or structural properties not easily obtained in binary cryptography.

# 7. Systems Engineering: Project Occam-7

Under the same research account, a substantial systems-engineering study — **USS Enterprise X / Project Occam-7** — applies rigorous first-principles modeling, digital-twin construction, and adversarial validation to the design of a reusable deep-space exploration vehicle. While thematically distinct from ternary computing, it shares the same methodological commitments: explicit assumptions, physical conservation laws as hard boundaries, adversarial defect discovery, and mission-level closure rather than subsystem optimism.

## Methodological Highlights

- Three levels of closure: mathematical, engineering (with margin), and mission (full dependency chain)
- Precursor autonomous robotic ISRU architecture that decouples crew departure from unverified surface production
- Explicit mass, energy, propellant, and state conservation enforced in the digital twin
- Hostile engineering: systematic search for model failures followed by remediation and re-evaluation
- Monte Carlo integrity and mission-state machine validation

The project demonstrates transferable research discipline: treating every important claim as something that must survive adversarial scrutiny and explicit prerequisite gating.

# 8. Ecosystem Structure & Governance

The research is organized as a multi-repository ecosystem with clear separation between normative specifications, interpretive documents, runtime implementations, tooling, and experimental domains.

## Domain Map

### Core Platform

t81-foundation, t81-vm, t81-lang, t81lib, t81-python, t81-examples, t81-benchmarks

### Specifications & Research

duotronic-whitepaper, duotronic-computing, t81-constraints, t81-roadmap, ternary-delta

### Model & Inference

ternary, llama.cpp (fork), ANGELA

### Hardware & Systems

t81-hardware, ternary-fabric, ternary-memory-research, ternary_gcc_plugin, ternary-tools

### Security & Crypto

trinity, trinity-pow, trinity-decrypt

### Documentation

t81-docs (central hub), ecosystem inventory, runtime contracts

## Governance Principles

- Normative specifications kept distinct from interpretive documents
- Deterministic interfaces preferred between research layers
- Explicit migration paths from binary-native systems
- Phase-based roadmap (P0 Alignment → P5 Public Release Readiness) with milestone matrices
- Adversarial testing and transparent defect registers

# 9. Roadmap Snapshot

### Phase 0 — Alignment & Baselines

Ecosystem constraint register, formal vs interpretive boundaries, baseline specs.

### Phase 1 — Core Stack Stabilization

Lock runtime semantics, stabilize VM, library, and language surfaces.

### Phase 2 — Tooling & Developer Experience

Docs unification, examples, onboarding paths, migration guides.

### Phase 3 — Hardware & Systems Prototyping

Simulation milestones, FPGA targets, memory research progression.

### Phase 4 — Security & Applied Domains

Formalize crypto evaluation plans, threat models, benchmark alignment.

### Phase 5 — Public Release Readiness

Release packaging, ecosystem coordination, contributor onboarding.

Longer-horizon directions include ternary-native ASICs, deeper quantum-inspired ideas, and the implications of deterministic, policy-governed substrates for advanced AI systems. Near-term emphasis remains on stabilizing the deterministic core and producing reproducible evidence.

# 10. Closing Statement

The T81 research program treats balanced ternary not as a nostalgic revival of historical machines, but as a contemporary engineering tool for domains that value determinism, auditability, and structural clarity. By combining a frozen ternary ISA, a policy-gated runtime, content-addressed decision objects, practical quantization paths for modern LLMs, and adversarial hardware/memory studies, the work aims to make ternary computing a measurable, falsifiable, and incrementally adoptable option rather than a purely theoretical curiosity.

All repositories are public. Contributions, independent replications, and critical evaluations are welcome.

**Primary Entry Points**

github.com/t81dev · github.com/t81dev/t81-foundation
github.com/t81dev/t81-docs · github.com/t81dev/t81-roadmap

This portfolio is derived from publicly available repository content as of October 2026.
It is intended as a research summary and does not constitute formal publication or claim of production readiness.